Case Study Work

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```
# 1. Load the original data
train_raw <- read.csv("data/data-train.csv")
test_raw  <- read.csv("data/data-test.csv")

train <- train_raw
test  <- test_raw

predictors <- c("St", "Re", "Fr")
raw_moments <- paste0("R_moment_", 1:4)
```

```
# 2. Check required columns
stopifnot(
  all(c(predictors, raw_moments) %in% names(train)),
  all(predictors %in% names(test))
)

# Check that the required columns are numeric and have no missing values.
# Fr = Inf is an intentional simulation setting, not a missing value.
stopifnot(
  all(vapply(train[c(predictors, raw_moments)], is.numeric, logical(1))),
  all(vapply(test[predictors], is.numeric, logical(1))),
  !anyNA(train[c(predictors, raw_moments)]),
  !anyNA(test[predictors])
)

# Re, St, and the raw moments must be finite.
stopifnot(
  all(is.finite(as.matrix(train[c("St", "Re", raw_moments)]))),
  all(is.finite(as.matrix(test[c("St", "Re")]))),
  all(train$Fr > 0),
```

```
  all(test$Fr > 0)
)
```

```
# 3. Convert raw moments into distribution summaries
m1 <- train$R_moment_1
m2 <- train$R_moment_2
m3 <- train$R_moment_3
m4 <- train$R_moment_4

train$mean_volume <- m1
train$variance_volume <- m2 - m1^2

# Stop and investigate if any variance is nonpositive.
if (any(train$variance_volume <= 0)) {
  stop("At least one calculated variance is nonpositive. Inspect those rows.")
}

train$sd_volume <- sqrt(train$variance_volume)

central_moment_3 <- m3 - 3 * m1 * m2 + 2 * m1^3
central_moment_4 <- m4 - 4 * m1 * m3 + 6 * m1^2 * m2 - 3 * m1^4

train$skewness_volume <- central_moment_3 / train$sd_volume^3
train$kurtosis_volume <- central_moment_4 / train$sd_volume^4

response_names <- c(
  "mean_volume", "variance_volume", "sd_volume",
  "skewness_volume", "kurtosis_volume"
)

stopifnot(
  all(is.finite(as.matrix(train[response_names])))
)
```

```
# 4. Print a compact preparation check
preparation_check <- data.frame(
  dataset = c("Training", "Test"),
  rows = c(nrow(train_raw), nrow(test_raw)),
  original_columns = c(ncol(train_raw), ncol(test_raw)),
  missing_entries = c(sum(is.na(train_raw)), sum(is.na(test_raw))),
  duplicate_input_settings = c(
    sum(duplicated(train_raw[predictors])),
```

```
    sum(duplicated(test_raw[predictors]))
  ),
  infinite_Fr_settings = c(
    sum(is.infinite(train_raw$Fr)),
    sum(is.infinite(test_raw$Fr))
  )
)

knitr::kable(preparation_check, caption = "Data preparation checks")
```

Table 1: Data preparation checks

dataset	rows	original_columns	missing_entries	duplicate_input_settings	infinite_Fr_settings
Training	89	7	0	0	30
Test	23	3	0	0	12

```
# Preview the predictors and calculated responses
knitr::kable(
  head(train[c(predictors, response_names)]),
  digits = 5,
  caption = "First six training observations"
)
```

Table 2: First six training observations

St	Re	Fr	mean_volume	variance_volume	sd_volume	skewness_volume	kurtosis_volume
0.10	224	0.052	0.00216	0.13035	0.36103	305.42800	93371.6555
3.00	224	0.052	0.00379	0.47041	0.68586	216.76225	47012.2515
0.70	224	Inf	0.00291	0.04349	0.20854	90.58974	8216.7778
0.05	90	Inf	0.06353	0.08662	0.29431	17.67980	420.2523
0.70	398	Inf	0.00037	0.00622	0.07889	257.58554	66372.7845
2.00	90	0.300	0.14780	1.98496	1.40888	12.64607	165.0999

```
# Summarize the calculated responses
summary(train[response_names])
```

```
  mean_volume       variance_volume        sd_volume         skewness_volume
 Min.   :0.000222   Min.   :1.000e-04   Min.   : 0.01006   Min.   : 11.97
 1st Qu.:0.002157   1st Qu.:2.447e-02   1st Qu.: 0.15643   1st Qu.: 72.55
```

```
Median :0.002958   Median :8.078e-02   Median : 0.28423   Median :110.12
Mean   :0.040394   Mean   :9.249e+01   Mean   : 3.65112   Mean   :162.81
3rd Qu.:0.087868   3rd Qu.:5.268e-01   3rd Qu.: 0.72579   3rd Qu.:269.54
Max.   :0.172340   Max.   :1.044e+03   Max.   :32.31526   Max.   :344.91
kurtosis_volume
Min.   :   150.5
1st Qu.:  5622.3
Median : 12158.7
Mean   : 39749.6
3rd Qu.: 72732.4
Max.   :132136.7
```